Particle Physics Homework 5

1). Show that for a scalar field theory
$$[\hat{H}, \hat{a}_{\vec{p}}] = -\omega_{\vec{p}} \hat{a}_{\vec{p}} .$$

2). Show that for a scalar field theory
$$[\hat{H}, \hat{P}] = 0 .$$
Does this make sense?

3). Calculate
$$\hat{H}(\hat{a}^{\dagger}_{\vec{p}} |0\rangle),$$
and justify your result based on previous question

4). Using the BCH formula:
$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{[\hat{A}, [\hat{A}, \hat{B}]]}{2!} + \frac{[\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]]}{3!} + \ldots$$
to calculate
$$e^{i\hat{H}t} \hat{a}_{\vec{p}} e^{-i\hat{H}t} .$$

5). Calculate
$$\langle 0 | \hat{\pi}_H(x^\mu) \hat{\pi}_H(y^\mu) | 0 \rangle$$